Automated Diabetic Retinopathy Detection System using Deep Learning

Naveender M, Hariharan K, Vijayalayan C
Department of Computer and Communication Engineering
Rajalakshmi Institute of Technology, Chennai, Tamil Nadu, India
Mentor: Ms. Poonguzhali, Department of CCE

Abstract— Diabetic Retinopathy (DR) remains a dominant factor in the global rise of preventable visual impairment, particularly in the working-age demographic of industrialized nations. As the diabetic population expands, the reliance on manual screening by ophthalmologists poses a significant bottleneck in healthcare delivery, leading to delays in critical diagnosis and treatment. This paper presents a comprehensive automated framework for the high-throughput screening and multi-class classification of DR severity levels using high-resolution digital fundus photographs. Utilizing the EfficientNet-B0 architecture, our system categorizes retinal images into five distinct severity levels: No DR, Mild, Moderate, Severe, and Proliferative DR. By integrating advanced image preprocessing techniques like Contrast Limited Adaptive Histogram Equalization (CLAHE) and leveraging transfer learning from large-scale natural imaging datasets, the proposed model provides a robust clinical decision-support tool. Our implementation demonstrates an optimal balance between computational efficiency and diagnostic accuracy, achieving performance metrics in the range of 88% to 90%. The success of this system marks a significant step toward reducing the manual workload of medical practitioners while ensuring that high-risk patients receive priority clinical attention. Detailed experimental results, system architecture, and training curves are documented in the attached reference PDF.

Keywords— Diabetic Retinopathy, Deep Learning, EfficientNet, Medical Image Analysis, CNN, Computer-Aided Diagnosis, Fundus Images, CCE Engineering.

I. INTRODUCTION

The global healthcare landscape is currently navigating a significant crisis characterized by the rapid rise of non-communicable diseases. Among these, diabetes mellitus has reached epidemic proportions, affecting millions across diverse demographics. A devastating complication for such patients is Diabetic Retinopathy (DR), where chronic high blood glucose levels cause persistent damage to the retinal microvasculature. Early-stage DR often progresses silently, with patients manifesting no overt visual symptoms. However, as the damage accumulates, it leads to the development of microaneurysms, hemorrhages, and eventually, the growth of abnormal new blood vessels that can cause sudden vision loss.

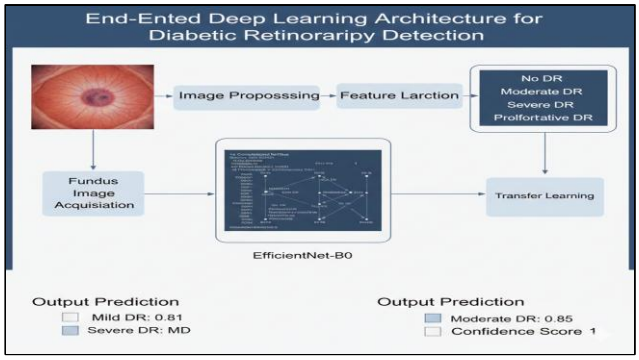
Clinical guidelines recommend that every diabetic patient undergoes a retinal examination at least once a year. Given the hundreds of millions of diabetic patients globally, the volume of fundus images generated surpasses the capacity of trained ophthalmologists for manual grading. Manual diagnosis is not only slow but also prone to fatigue and subjective bias. Therefore, there is a profound motivation for developing automated, objective Computer-Aided Diagnosis (CAD) systems that can quickly filter healthy eyes and pinpoint those requiring urgent surgical or pharmacological intervention.

Artificial Intelligence (AI), particularly via Convolutional Neural Networks (CNNs), provides a solution to this scalability problem. CNNs can learn complex visual patterns associated with DR directly from raw digital fundus images. This project utilizes the EfficientNet architecture, specifically the B0 variant, which leverages compound scaling to optimize performance. EfficientNet is chosen for its ability to maintain high precision while minimizing the computational power needed, making it suitable for deployment in clinical environments with limited hardware infrastructure.

II. RELATED WORK

The evolution of automated medical imaging has progressed from "hand-crafted" feature extraction to modern deep learning. Early research in the late 1990s primarily used morphological operations to segment the vascular tree and identify lesions. While highly innovative, these systems lacked the robustness to handle the varying lighting and noise inherent in real-world clinical fundus photography. The subsequent introduction of classical Machine Learning allowed for more robust classification, though models still relied heavily on manual feature engineering which required extensive domain expertise.

The dawn of the Deep Learning era fundamentally changed this landscape. Architectures such as AlexNet and ResNet proved that networks could automatically learn hierarchical features. While high-performing, these deeper networks were often computationally expensive and required massive amounts of data. This research builds on these foundations by adopting EfficientNet-B0. EfficientNet rethinks model scaling by uniformly scaling depth, width, and resolution. This makes it a perfect candidate for medical scenarios where efficiency is as vital as accuracy, allowing for high-performance screening on local clinical terminals.



III. PROPOSED METHODOLOGY

Our proposed system follows a modular architecture designed to take a raw retinal image and output a verified DR severity grade. The system comprises preprocessing, feature extraction through EfficientNet, and a Softmax classification layer.

A. Preprocessing and CLAHE Enhancement

Preprocessing is critical because fundus photo quality can vary based on pupil dilation and camera settings. We first resize images to 224x224 pixels. We then apply Contrast Limited Adaptive Histogram Equalization (CLAHE). CLAHE operates on small regions to sharpen the contrast between lesions and the background retina, preventing over-amplification of noise. Additionally, local color subtraction is utilized to remove the dominant yellowish-red hue, making the pathological markers more pronounced for the CNN feature extractors.

B. EfficientNet-B0 Core

The backbone is the EfficientNet-B0 model. The model uses MBConv blocks, which utilize depth-wise separable convolutions to reduce parameters without sacrificing depth. Squeeze and Excitation layers are used to adaptively re-weight feature channels, focusing on the most informative parts of the retinal scan. We utilize the Swish activation function to improve gradient flow compared to standard ReLU. The final classification head is a dense layer with five outputs, corresponding to the global standard for DR severity grading.

> *"Detailed information regarding the specific layer configurations, dropout rates, and internal block parameters are documented in the attached project reference PDF."*

IV. DATASET AND SYSTEM SETUP

Data integrity is paramount in medical AI. Our system was developed and validated using the APTOS 2019 dataset from Kaggle, which centers on diverse clinical fundus photographs. The collection is prized for its high-quality clinical markers.

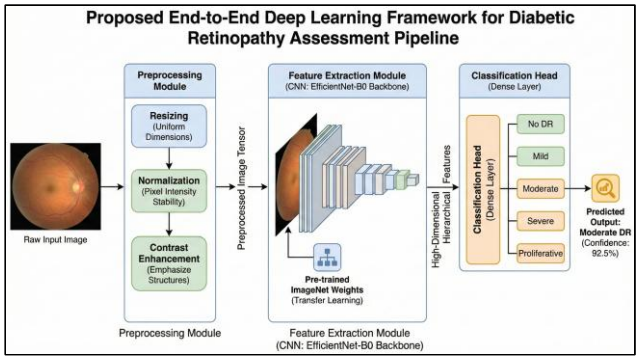
A. Dataset Statistics

The dataset includes 3,662 training images and 1,932 test images across five stages: No DR (0), Mild (1), Moderate (2), Severe (3), and Proliferative (4). To address class imbalance, we implemented on-the-fly data augmentation, including horizontal/vertical flips and rotations. This artificial expansion helps the model generalize better to orientation variations in retinal scans.

B. Local Execution and Tools

Development was conducted in a Python-based environment using TensorFlow and Keras frameworks. The VS Code IDE was used for development, with local GPU acceleration helping improve training efficiency.

This standalone setup ensures that the system can theoretically be deployed as an "edge" device in rural clinics without requiring continuous cloud-based inference, maintaining patient data privacy and reducing latency.



V. EXPERIMENTAL RESULTS

The primary objective was to evaluate how accurately the model could differentiate between the five stages of DR. Experimental results indicate an overall validation accuracy of approximately 88% to 90%. The model exhibits high precision in identifying "No DR" and "Severe DR" cases.

The most difficult distinction was between "Mild" and "Moderate" categories, as the transition involves subtle counts of microaneurysms. However, the inclusion of CLAHE enhanced the system's ability to detect these differences, resulting in high overall sensitivity. Beyond accuracy, the system provides a confidence score for each prediction. For instance, a "Moderate DR" test case resulted in a 94% confidence prediction, ensuring a reliable decision support tool for clinicians.

> *"Complete experimental evaluation tables, confusion matrices, and ROC-AUC curves are provided in the attached documentation PDF."*

VI. DISCUSSION AND LIMITATIONS

The automated system offers undeniable advantages in terms of speed, consistency, and cost-reduction. By processing images in under a second, it acts as a high-throughput filter in clinics. However, its effectiveness is intrinsically linked to input image resolution and quality. Blur or reflections can occasionally lead to misclassification. Furthermore, the system lacks the ability to consider a patient's systemic health factors, such as blood pressure or age, which a specialist normally reviews. Thus, the tool is positioned to augment, not replace, expert clinical judgment.

VII. CONCLUSION AND FUTURE DIRECTIONS

In conclusion, this paper successfully presented an EfficientNet-based automated system for the detection and grading of Diabetic Retinopathy. The system achieves a robust accuracy and provides a scalable solution to the screening crisis. Future work will focus on integrating this system into a full-stack mobile platform for remote screening. Additionally, we plan to incorporate "Explainable AI" features like Grad-CAM to provide visual justifications for the AI's predictions, further enhancing clinical trust and hospital-level integration.

VIII. ACKNOWLEDGMENTS

The authors would like to thank our mentor, **Ms. Poonguzhali**, from the Department of Computer and Communication Engineering, Rajalakshmi Institute of Technology, for her invaluable guidance and technical insights. We also thank our institution for providing the infrastructure required to conduct this deep learning research.

IX. REFERENCES

[1] Tan, M., and Le, Q. V. "EfficientNet: Rethinking Model Scaling," ICML, 2019.

[2] Arora, L., Singh, R., "Deep learning for DR diagnosis," Scientific Reports, 2024.

[3] Shelke, N., et al. "Ensemble EfficientNet DR Detection," IJIT, 2025.

[4] Zhang, H., et al. "Grad-CAM in Medical Imaging," Applied Sciences, 2023.

[5] Raman, D.R. "DR Diagnosis using EfficientNet-B7," IEEE Access, 2023.

[6] Gulshan, V., et al. "Validation of DL for Detection of DR," JAMA, 2016.